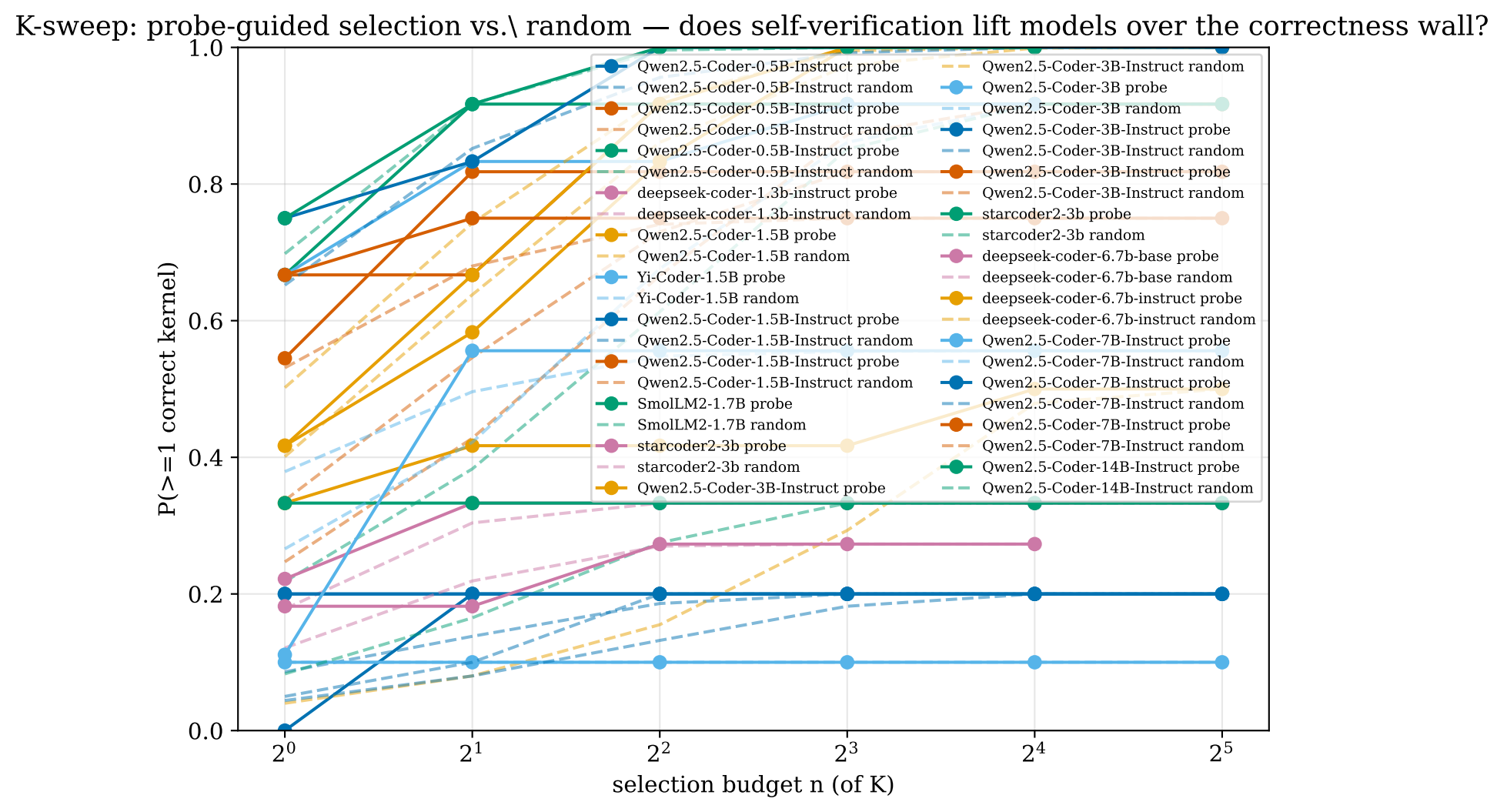

K-sweep: probe-guided selection vs.\ random — does self-verification lift models over the correctness wall?
P(>=1 correct kernel)
selection budget n (of K)
1.0
0.8
0.6
0.4
0.2
0.0
2^0
2^1
2^2
2^3
2^4
2^5
Qwen2.5-Coder-0.5B-Instruct probe
Qwen2.5-Coder-0.5B-Instruct random
Qwen2.5-Coder-0.5B-Instruct probe
Qwen2.5-Coder-0.5B-Instruct random
Qwen2.5-Coder-0.5B-Instruct probe
Qwen2.5-Coder-0.5B-Instruct random
deepseek-coder-1.3b-instruct probe
deepseek-coder-1.3b-instruct random
Qwen2.5-Coder-1.5B probe
Qwen2.5-Coder-1.5B random
Yi-Coder-1.5B probe
Yi-Coder-1.5B random
Qwen2.5-Coder-1.5B-Instruct probe
Qwen2.5-Coder-1.5B-Instruct random
Qwen2.5-Coder-1.5B-Instruct probe
Qwen2.5-Coder-1.5B-Instruct random
SmolLM2-1.7B probe
SmolLM2-1.7B random
starcoder2-3b probe
starcoder2-3b random
Qwen2.5-Coder-3B-Instruct probe
Qwen2.5-Coder-3B-Instruct random
Qwen2.5-Coder-3B probe
Qwen2.5-Coder-3B random
Qwen2.5-Coder-3B-Instruct probe
Qwen2.5-Coder-3B-Instruct random
Qwen2.5-Coder-3B-Instruct probe
Qwen2.5-Coder-3B-Instruct random
starcoder2-3b probe
starcoder2-3b random
deepseek-coder-6.7b-base probe
deepseek-coder-6.7b-base random
deepseek-coder-6.7b-instruct probe
deepseek-coder-6.7b-instruct random
Qwen2.5-Coder-7B-Instruct probe
Qwen2.5-Coder-7B-Instruct random
Qwen2.5-Coder-7B-Instruct probe
Qwen2.5-Coder-7B-Instruct random
Qwen2.5-Coder-7B-Instruct probe
Qwen2.5-Coder-7B-Instruct random
Qwen2.5-Coder-14B-Instruct probe
Qwen2.5-Coder-14B-Instruct random